

Sergei Chudov

Software Engineer

chudov42@gmail.com · linkedin.com/in/chudov-sergei · github.com/WindowGenerator · telegram · +374 91 186 658 · Yerevan, Armenia

Summary

Software Engineer with **7+ years** of experience building backend services, data pipelines, and production systems. Strong expertise in Python, Kubernetes, microservices, and DevOps practices. Experienced in taking services from requirements through implementation, deployment, monitoring, and production support. Contributed to open source projects including ruff, aio-pika, and chat-langchain. Background includes bioinformatics, security systems, and cloud-native infrastructure.

Experience

Lead Software Engineer — *BostonGene, Yerevan*

Mar 2025 – Present

- Develop workflow tools for cross-functional teams, owning requirements, backend implementation, deployment, and production support.
- Maintain reliability of Kubernetes-based production services through monitoring, operational support, and load-related adjustments.
- Led design and delivery of a backend billing service for annotation workflows.
- Wrote `netfs_unlker` (Rust) to recover a locked NetApp DB, saving **2 weeks of work** across multiple teams.
- Delivered an initial production Python library with tests and CI/CD in **4 days**, including a small language for converting patient parameters and relationships into machine-readable format.
- Introduced team-wide testing, linting, and formatting standards, improving development consistency and reducing avoidable rework.
- **Stack:** Python · FastAPI · asyncio · Airflow · Luigi · TypeScript · React · Docker · Kubernetes · Helm · Concourse · GitLab CI · Vault · Prometheus · Kibana/Vector · PostgreSQL · RabbitMQ

Senior Software Engineer — *BostonGene, Yerevan*

Jun 2022 – Mar 2025

- Productionized bioinformatics algorithms and data pipelines by implementing backend services, frontend changes, and pipeline integrations.
- Centralized logging for **10+ services** using Vector and Kibana, improving incident investigation across backend teams.
- Designed migration from static AWS keys to IRSA and Vault, reducing credential leakage risk and aligning services with access policies.
- Built shared integration libraries and an internal Python SDK with documentation and CI/CD, improving reuse across backend services.
- Improved reliability of Kubernetes-based pipeline execution by refining Luigi logging, scheduler behavior, and worker failure handling.
- Wrote Helm charts and configured CI/CD pipelines with Concourse and GitLab CI for production projects.
- **Stack:** Python · FastAPI · SQLAlchemy · asyncio · Luigi · Airflow · TypeScript · React · Redux · Docker · Kubernetes · Helm · Concourse · GitLab CI · Vault · Prometheus · Kibana/Vector · PostgreSQL · Neo4j · Redis · RabbitMQ · Kafka

Software Engineer — *USSC Ltd., Yekaterinburg*

Oct 2019 – Mar 2022

- Built backend components for a system detecting and preventing information-security incidents in industrial networks.
- Refactored authorization logic to simplify permission handling and reduce recurring authorization defects.
- Rewrote complex master-master interaction functionality and covered it with tests, reducing recurring failures.
- Created an OpenAPI-to-TypeScript client generator, reducing generated client memory usage and improving client-side method signatures.
- Built MySQL/PostgreSQL backup service and Elasticsearch index-rotation tooling.
- Designed Python microservices, optimized SQL in MySQL/PostgreSQL, wrote Docker images, and maintained unit and integration tests.
- **Stack:** Python · FastAPI · aiohttp · Flask · SQLAlchemy · TypeScript · Angular · Docker · Jenkins · PostgreSQL · MySQL · Elasticsearch · RabbitMQ

- Prototyped a real-time face identification system from video camera data using YOLOv3.
- Trained additional YOLOv3 model components, optimized inference on low-power hardware, and evaluated object-recognition models.
- **Stack:** Python · TensorFlow · Keras · scikit-learn

Skills

Languages | Python · Go · Rust · TypeScript · SQL

Backend | FastAPI · aiohttp · Flask · asyncio · SQLAlchemy · microservices

Infrastructure | Docker · Kubernetes · Helm · Linux · CI/CD (Concourse, GitLab CI, Jenkins) · Prometheus · Kibana/Vector · Vault

Data | PostgreSQL · MySQL · Redis · RabbitMQ · Kafka · Elasticsearch · Neo4j · Luigi · Airflow · S3

Open Source & Projects

- Contributions: `aio-pika` · `ruff` · `chat-langchain`
- `starlette-responses-kit` — custom Starlette response types
- `netfs_unlker` — Rust CLI to unlock `fcntl` shared locks on NetApp volumes

Education

Ural State Technical University — B.Eng. Design & Technology of Electronic Facilities

2015 – 2020

Certifications & Courses

Coursera: Algorithms Part I (2020) · Intro to Machine Learning (2021)

Languages

Russian — native | English — professional working